PRESENTATION OF A COMPLETE LOCAL RING OF A LOCALLY NOETHERIAN SCHEME AS A QUOTIENT OF A POWER SERIES RING, AND WHEN IT IS A K-ALGEBRA ISOMORPHISM

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Two of the author's footnotes belong to the counted proof span and are reproduced verbatim: the one explaining the mixed characteristic form of the Cohen structure theorem, and the aside pointing to the computation modulo y^2.

Editorial note. The counted spans carry 7 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-complete-local-ring-structure-as-algebra is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Cohen structure theorem for complete local rings. Algebra, Theorem algebra-theorem-cohen-structure-theorem, invoked at proof lines 4060-4067. Supplies the existence of the presentation by a quotient of a power series ring.
- Lifting the residue field along a separable extension. More on Algebra, Lemma more-algebra-lemma-lift-residue-field, invoked at proof lines 4081-4084. Supplies the k-algebra homomorphism kappa to the completion when kappa/k is separable.
- Completeness of a finitely generated module and the surjectivity criterion for adically complete maps. Algebra, Lemmas algebra-lemma-hathat-finitely-generated and algebra-lemma-completion-generalities, invoked at proof lines 4091-4096. Supplies that the constructed map of complete rings is surjective once it is surjective modulo the augmentation ideal.

Source reference. Original lemma 0315: sources/stacks-chow/tags.txt, source lines 3972–3972. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 05GG: sources/stacks-chow/tags.txt, source lines 7060–7060. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original theorem 032A: sources/stacks-chow/tags.txt, source lines 4013–4013. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0C34: sources/stacks-chow/tags.txt, source lines 15179–15179. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Let X be a locally Noetherian scheme over k. Let $x \in X$ be a point with residue field κ. There is an isomorphism*

$$\kappa[[x_1, \ldots, x_n]]/I \longrightarrow \mathcal{O}^\wedge_{X,x} \tag{0.1.1}$$

inducing the identity on residue fields. In general we cannot choose (0.1.1) to be a k-algebra isomorphism. However, if the extension κ/k is separable, then we can choose (0.1.1) to be an isomorphism of k-algebras.

Proof. The existence of the isomorphism is an immediate consequence of the Cohen structure theorem[1] (Algebra, Theorem 032A).

Let p be an odd prime number, let $k = \mathbf{F}_p(t)$, and $A = k[x, y]/(y^2 + x^p - t)$. Then the completion $A^\wedge$ of A in the maximal ideal $\mathfrak{m} = (y)$ is isomorphic to $k(t^{1/p})[[z]]$ as a ring but not as a k-algebra. The reason is that $A^\wedge$ does not contain an element whose pth power is t (as the reader can see by computing modulo y^2). This also shows that any isomorphism (0.1.1) cannot be a k-algebra isomorphism.

If κ/k is separable, then there is a k-algebra homomorphism $\kappa \to \mathcal{O}^\wedge_{X,x}$ inducing the identity on residue fields by More on Algebra, Lemma 0C34. Let $f_1, \ldots, f_n \in \mathfrak{m}_x$ be generators. Consider the map

$$\kappa[[x_1, \ldots, x_n]] \longrightarrow \mathcal{O}^\wedge_{X,x}, \quad x_i \longmapsto f_i$$

Since both sides are $(x_1, \ldots, x_n)$-adically complete (the right hand side by Algebra, Lemmas 05GG) this map is surjective by Algebra, Lemma 0315 as it is surjective modulo $(x_1, \ldots, x_n)$ by construction. □

[1] Note that if κ has characteristic p, then the theorem just says we get a surjection $\Lambda[[x_1, \ldots, x_n]] \to \mathcal{O}^\wedge_{X,x}$ where Λ is a Cohen ring for κ. But of course in this case the map factors through $\Lambda/p\Lambda[[x_1, \ldots, x_n]]$ and $\Lambda/p\Lambda = \kappa$.